

List Induction (Part 2: Complex Properties & Case Analysis)

Lecture Notes

Course: Computer-Assisted Theorem Proving (7CCSACTP)

Definitions & Goal Statement

Functions:

- $\text{Max_list } [] = 0, \quad \text{Max_list } (x \# xs) = \max x (\text{Max_list } xs)$
- $\text{List_set } [] = \emptyset, \quad \text{List_set } (x \# xs) = \text{insert } x (\text{List_set } xs)$

Goal: Prove $x \in \text{List_set } xs \implies x \leq \text{Max_list } xs$ by induction on xs .

Base Case ($xs = []$)

Assumptions & Goal:

- Assumption: $x \in \text{List_set } []$
- Goal: $x \leq \text{Max_list } []$

Proof Steps:

1. $\text{Max_list } [] = 0$ (by *Max_list.simps*)
2. $x \in \emptyset$ (from base assumption & *List_set.simps*)
3. False (using *lem3 [2, lem1]*)
4. $x \leq \text{Max_list } []$ (using *lem2 [OF 3]*)

Step Case ($xs \rightarrow y \# ys$)

Induction Hypothesis (I.H.): $x \in \text{List_set } ys \implies x \leq \text{Max_list } ys$

Premise: $x \in \text{List_set } (y \# ys)$ **Goal:** $x \leq \text{Max_list } (y \# ys)$

Proof by Case Analysis on $x = y$:

Case 1 ($x = y$):

1. $y \leq \text{Max_list } (y \# ys)$ (from *lem1*)
2. $x \leq \text{Max_list } (y \# ys)$ (by *subst case assump x = y*)

Case 2 ($x \neq y$):

1. $x \in \text{List_set } ys$ (from *case assump* & *lem2*)
2. $x \leq \text{Max_list } ys$ (from I.H. [OF 1])
3. $\text{Max_list } ys \leq \text{Max_list } (y \# ys)$ (using *lem3*)
4. $x \leq \text{Max_list } (y \# ys)$ (using *lem4 [OF 2 3]*) ✓

Auxiliary Lemmas Used in Proof

- *lem1*: $a \leq \text{Max_list } (a \# xs)$
- *lem2*: $a \in \text{List_set } (b \# bs) \implies a = b \vee a \in \text{List_set } bs$
- *lem3*: $\text{Max_list } xs \leq \text{Max_list } (x \# xs)$
- *lem4* (Transitivity): $a \leq b \implies b \leq c \implies a \leq c$